

INTERNATIONAL STANDARD



Multimedia systems and equipment – Multimedia e-publishing and e-book technologies – Printing specification of texture map for auditory presentation of printed texts



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Multimedia systems and equipment – Multimedia e-publishing and e-book technologies – Printing specification of texture map for auditory presentation of printed texts

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**MULTIMEDIA SYSTEMS AND EQUIPMENT –
MULTIMEDIA E-PUBLISHING AND E-BOOK TECHNOLOGIES –
PRINTING SPECIFICATION OF TEXTURE MAP FOR AUDITORY
PRESENTATION OF PRINTED TEXTS**

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The text of this standard is based on the following documents:

CDV	Report on voting
100/2292/CDV	100/2400/RVC

Full information on the voting for the approval of this standard can be found in the report on voting indicated in the above table.

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MULTIMEDIA SYSTEMS AND EQUIPMENT – MULTIMEDIA E-PUBLISHING AND E-BOOK TECHNOLOGIES – PRINTING SPECIFICATION OF TEXTURE MAP FOR AUDITORY PRESENTATION OF PRINTED TEXTS

1 Scope

The texture map for auditory presentation of printed texts is printed on paper or shown on display devices. This International Standard specifies the printing quality of this texture map on paper.

In order to ensure an interoperability of the texture map specified in IEC 62665, a printing specification based on the quality of this International Standard should be employed.

2 Normative reference

The following documents, in whole or in part, are normatively referenced in this document and are indispensable for its application. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 62665, *Multimedia systems and equipment – Multimedia e-publishing and e-books technologies – Texture map for auditory presentation of printed texts*

3 Terms and definitions

For the purposes of this document, the following terms and definitions apply.

3.1

texture map

two dimensional cell patterns which include alignment lines and a data matrix

[SOURCE: IEC 62665, 2.1, modified – Definition has been clarified.]

3.2

data matrix

two dimensional cell patterns generated from text data compression and error correction encoding

3.3

cell

minimum square dot which constitutes a data matrix

3.4

alignment line

border line which encloses a texture map

Note 1 to entry: An alignment line consists of solid lines and tick marks. The solid lines have a cell's width and may partially be interrupted.

Note 2 to entry: For additional information on alignment lines, see IEC 62665.

3.5

tick mark

positioning mark which intersects perpendicularly with the solid lines and is given at equal intervals on the solid lines of an alignment line

Note 1 to entry: A tick mark indicates the unit boundary and has a cell's width.

3.6

quiet zone

non-printing area outside of a texture map

3.7

unit

square area which consists of 11×11 cells in a texture map

4 Printing specification

4.1 General

Texture maps generated by a system are specified by IEC 62665 and a printing quality of the texture maps has to be standardized to ensure interoperability of the printed texture maps.

Texture maps should be printed on paper with a resolution of ≥ 600 dpi (dots per inch, $\geq 23,622$ dot/mm). In order to minimize specular reflection, a mat coated paper or a low reflection paper should be used.

4.2 Size of a texture map

Four sizes of a texture map are specified by IEC 62665 and they have their dimensions shown in Table 1.

Table 1 – Sizes of a texture map

Size	Number of cells	Number of units	Dimensions at printing mm
XS	40×40	3×3	$6,8 \times 6,8$
S	73×73	6×6	$12,4 \times 12,4$
M	106×106	9×9	$17,9 \times 17,9$
L	117×117	10×10	$19,8 \times 19,8$

4.3 Printing quality of a texture map

4.3.1 Printing quality

Printing quality of a texture map is defined by the average Q of values of the following quality measures and indicated with the symbol A, B, C, D or F, as shown in Table 2.

The following aspects define the printing quality:

- print contrast;
- symmetry of cell pattern;
- squareness;
- size accuracy;
- tick mark identification;
- quiet zone sufficiency;

- g) uselessness of error correction, and
- h) decodability.

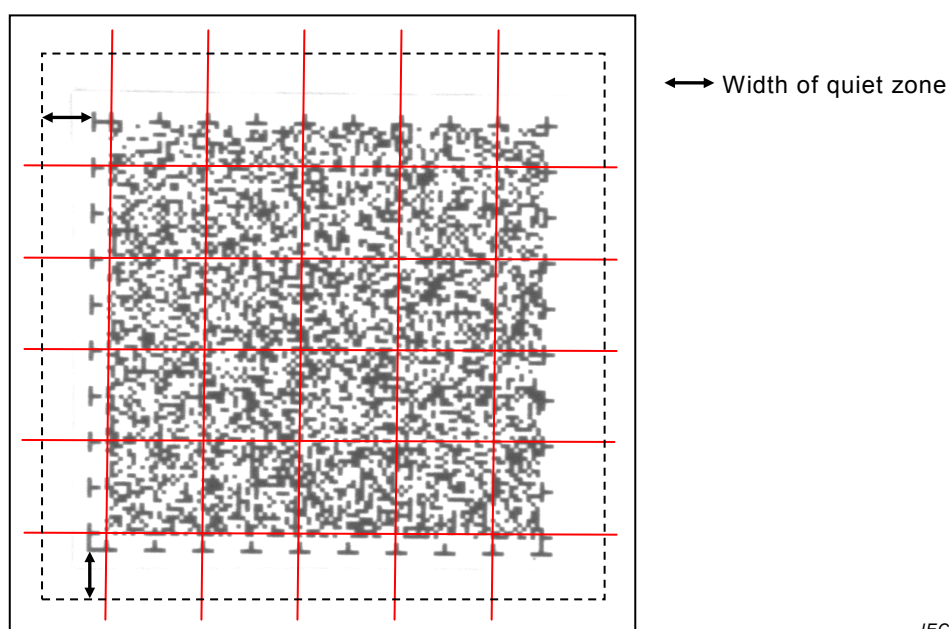
Table 2 – Printing quality of a texture map

Q (average of the values of quality measures a) through h))	Printing quality of a texture map
$3,4 \leq Q \leq 4,0$	A
$2,6 \leq Q < 3,4$	B
$1,8 \leq Q < 2,6$	C
$1,0 \leq Q < 1,8$	D
$Q < 1,0$	F

4.3.2 Sampling line

The values of quality measures a) through d) are defined by scanning cells on 10 sampling lines (5 horizontal and 5 vertical lines). A sampling line connects the corresponding tick marks on the opposite alignment lines, as shown in Figure 1. For the size XS texture map, every tick mark is connected by a sampling line. For size S, M and L texture maps, every other tick mark is connected by a sampling line.

The values of quality measures a) through d) are defined by scanning cells on 10 sampling lines (5 horizontal and 5 vertical lines). A sampling line connects the corresponding tick marks on the opposite alignment lines as shown in Figure 1. For size XS texture map, every tick mark is connected by a sampling line. For size S, M and L texture maps, every other tick mark is connected by a sampling line.



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Figure 1 – Sampling lines in case of size M texture map

4.3.3 Quality measure

Each quality measure is defined as follows:

a) print contrast

For the reflectance pattern of white and black cells on scanning the sampling lines, using

$RL_{\min}$ minimum value of reflectance of white cells,

RL_{ave} average value of reflectance of white cells, and

$RD_{\max}$ maximum value of reflectance of black cells,

RC is defined as

$$RC = (RL_{\min} - RD_{\max}) / RL_{\text{ave}}. \quad (1)$$

The value of the print contrast is determined in accordance with the range of RC , as shown in Table 3.

Table 3 – Value of print contrast

RC	Value of print contrast
$0,90 \leq RC$	4
$0,85 \leq RC < 0,90$	3
$0,80 \leq RC < 0,85$	2
$0,75 \leq RC < 0,80$	1
$RC < 0,75$	0

b) symmetry of cell pattern

For the reflectance pattern of white and black cells on scanning the sampling lines, using

RL_{ave} average value of reflectance of white cells, and

RD_{ave} average value of reflectance of black cells,

the threshold RT (see Figure 2) is defined as

$$RT = (RL_{\text{ave}} + RD_{\text{ave}}) / 2. \quad (2)$$

Using the maximum value $WL_{\max}$ and minimum values $WL_{\min}$ of

WL width of the reflectance pattern of 4 connected white cells (interval between the leading and trailing points crossing RT),

and the maximum value $WD_{\max}$ and minimum values $WD_{\min}$ of

WD width of the reflectance pattern of 4 connected black cells (interval between the leading and trailing points crossing the RT),

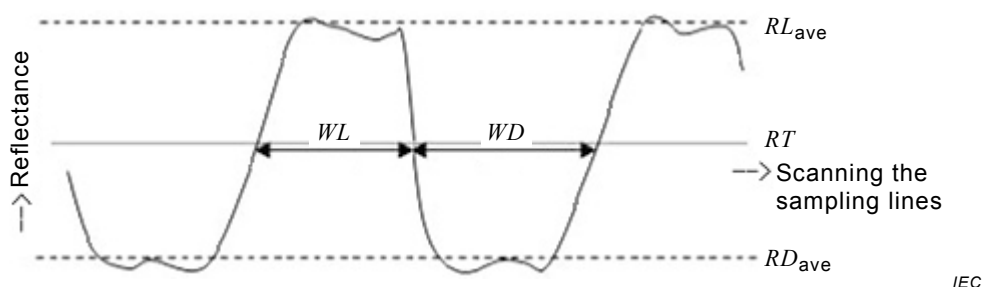


Figure 2 – Reflectance pattern of cells on scanning the sampling lines

WS is defined as

$$WS = |WD_{\max} - WL_{\min}| / WD_{\max} \text{ (or } |WD_{\min} - WL_{\max}| / WL_{\max}) \quad (3)$$

The value of the symmetry of cell pattern is determined in accordance with the range WS as shown in Table 4.

Table 4 – Value of symmetry of cell pattern

WS	Value of symmetry of cell pattern
$0,02 \geq WS$	4
$0,05 \geq WS > 0,02$	3
$0,10 \geq WS > 0,05$	2
$0,15 \geq WS > 0,10$	1
$WS > 0,15$	0

c) squareness

Using the average NX_{ave} of

NX number of cells between end points of tick marks on horizontal sampling lines (see Figure 3),

and the average NY_{ave} of

NY number of cells between end points of tick marks on vertical sampling lines,

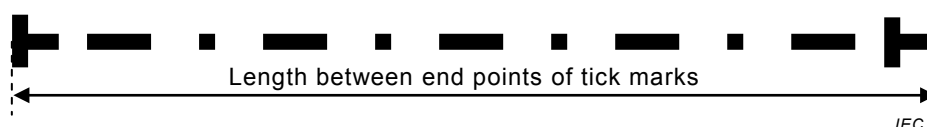


Figure 3 – Length between end points of tick marks on a horizontal sampling line

NQ is defined as

$$NQ = |NX_{\text{ave}} - NY_{\text{ave}}| / (NX_{\text{ave}} + NY_{\text{ave}}) / 2. \quad (4)$$

The value of squareness is determined in accordance with the range of NQ as shown in Table 5.

Table 5 – Value of squareness

NQ	Value of squareness
$0,005 \geq NQ$	4
$0,010 \geq NQ > 0,005$	3
$0,020 \geq NQ > 0,010$	2
$0,030 \geq NQ > 0,020$	1
$NQ > 0,030$	0

d) size accuracy

Using

$(NX_{\text{ave}} + NY_{\text{ave}}) / 2$ average number of cells between end points of tick marks on a sampling line, and

NE standard number of cells between end points of tick marks on a sampling line (see Figure 1),

NN (normalized difference between them) is defined as

$$NN = |NE - (NX_{ave} + NY_{ave}) / 2| / NE. \quad (5)$$

The value of size accuracy is determined in accordance with the range of NN as shown in Table 6.

Table 6 – Value of size accuracy

NN	Value of size accuracy
$0,005 \geq NN$	4
$0,010 \geq NN > 0,005$	3
$0,020 \geq NN > 0,010$	2
$0,030 \geq NN > 0,020$	1
$NN > 0,030$	0

e) tick mark identification

Using

NI number of identified tick marks, and

NS standard number of tick marks,

ND is defined as

$$ND = |NS - NI|. \quad (6)$$

The value of tick mark identification is determined in accordance with the value of ND as shown in Table 7.

Table 7 – Value of tick mark identification

ND	Value of tick mark identification
$ND = 0$	4
$ND \geq 1$	0

f) quiet zone sufficiency

Using

M horizontal and vertical width of the quiet zone,

the value of quiet zone sufficiency is determined in accordance with the value of M as shown in Table 8.

Table 8 – Value of quiet zone sufficiency

M mm	Value of quiet zone sufficiency
$M \geq 4$	4
$M < 4$	0

g) uselessness of error correction

Using

NC number of error correction enforcements in text decoding of a texture map,

the value of uselessness of error correction is determined in accordance with the range of NC as shown in Table 9.

Table 9 – Value of uselessness of error correction

NC	Value of uselessness of error correction
$5 \geq NC$	4
$10 \geq NC > 5$	3
$15 \geq NC > 10$	2
$20 \geq NC > 15$	1
$NC > 20$	0

h) decodability

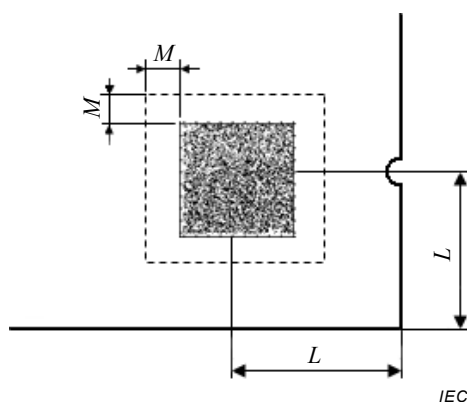
The value of decodability is determined in accordance with the result of the text decoding of a texture map as shown in Table 10.

Table 10 – Value of decodability

Result of text decoding	Value of decodability
Success	4
Failure	0

4.4 Location of a texture map

A texture map shall be printed with its center located at L ($25 \pm 0,5$) mm apart from each edges of paper as shown in Figure 4. Outside the texture map, the quiet zone shall be located with its horizontal and vertical width M of 4 mm or more.

**Figure 4 – Location of a texture map****4.5 Location of a notch**

A notch is required so that blind or vision impaired people may recognize a texture map on its paper. If a texture map is printed on a single side of the paper, a single notch shall be located, as shown in Figure 5, at an edge of the paper on which a texture map is printed. A notch shall have its diameter D of ($5 \pm 0,5$) mm.

If a texture map is printed on both sides of a paper, double notches shall be located, as shown in Figure 6, at the same edge of the paper. The distance d between notches shall be ($9 \pm 1,0$) mm.

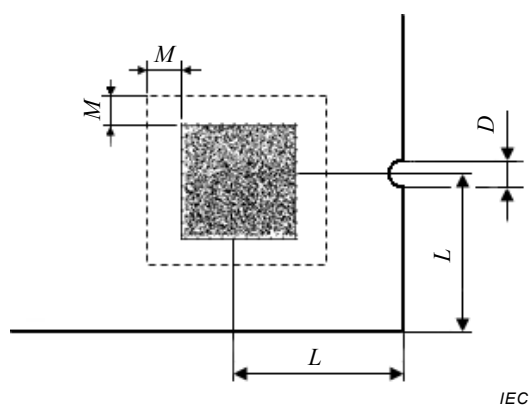


Figure 5 – Location of a single notch

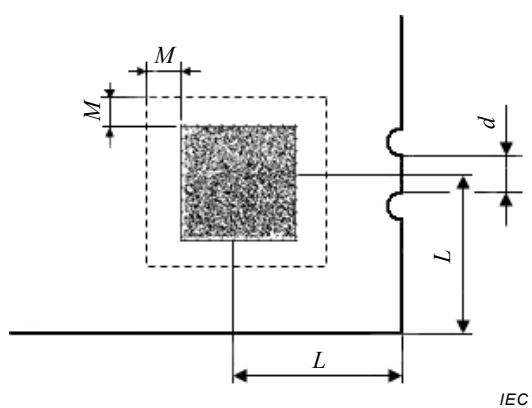


Figure 6 – Location of double notches

Annex A

(informative)

Examples of visible quality of a texture map

Visible quality of a texture map depends on the constitution of cells that will affect the

- a) print contrast,
- b) symmetry of cell pattern,
- c) squareness,
- d) size accuracy, and
- e) tick mark identification,

Some examples of visible quality are shown in Figure A.1, Figure A.2, Figure A.3 and Figure A.4.

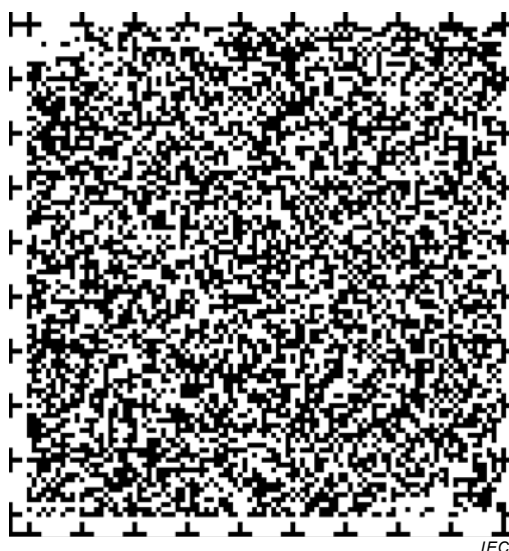


Figure A.1 – Texture map with high printing quality

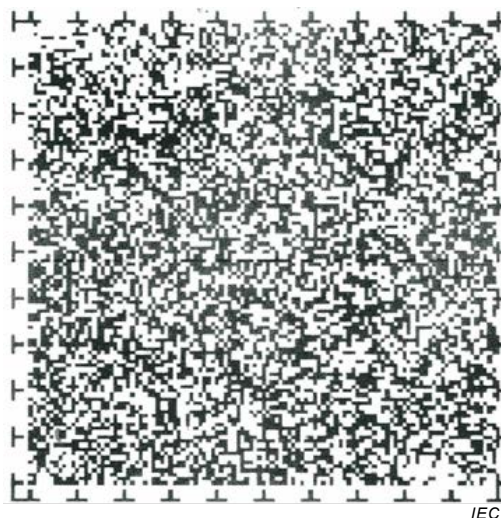


Figure A.2 – Texture map including low contrast cells and non-square cells

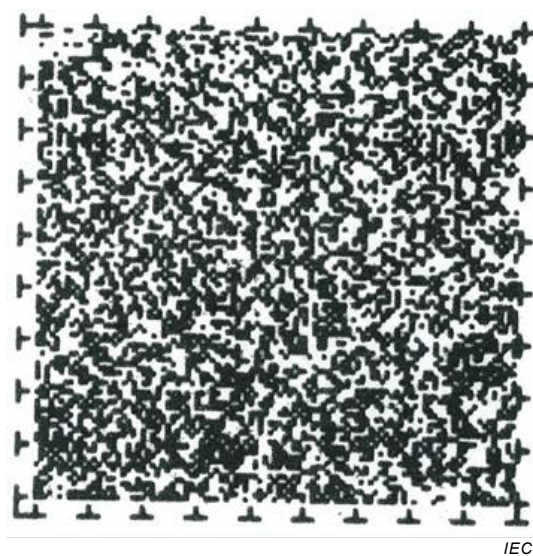


Figure A.3 – Texture map with low sharpness

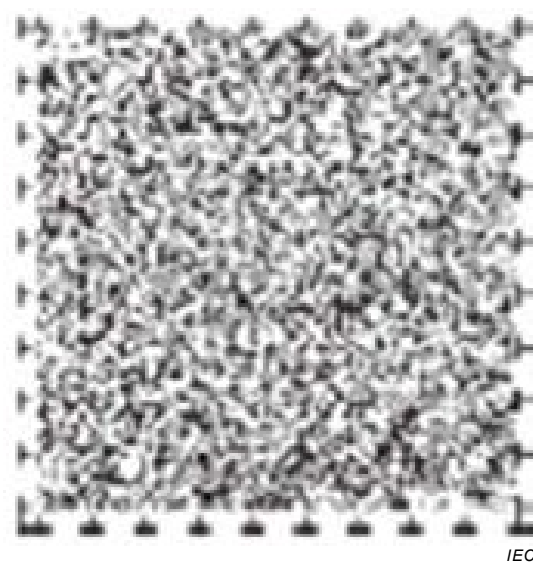
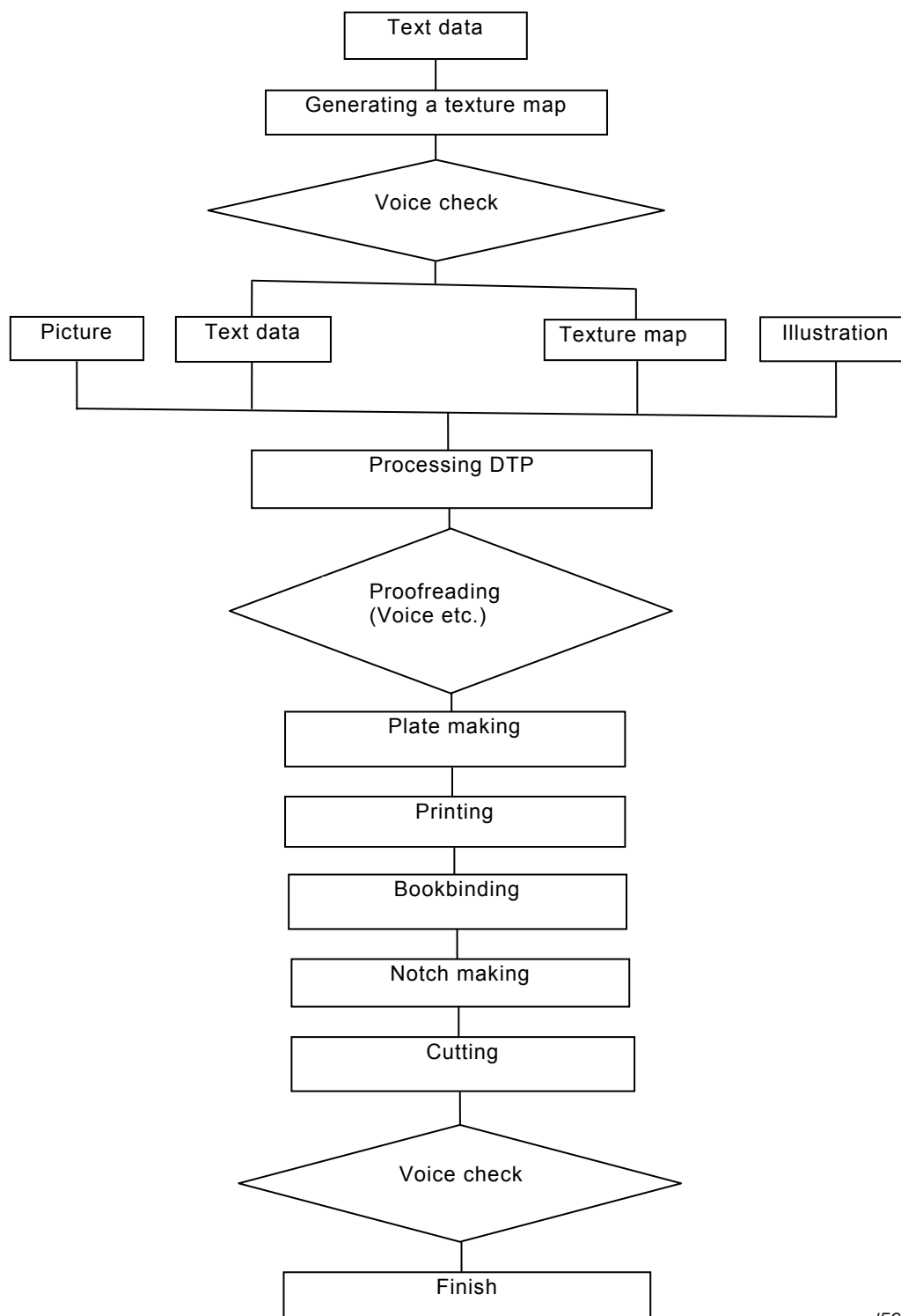


Figure A.4 – Texture map processed by non-preserving data compression

Annex B (informative)

Creating process of printed matter including a texture map

A typical flow of creating process of printed matter including a texture map is shown in Figure B.1.



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Figure B.1 – Creating process of printed matter including a texture map

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